

Supplier databases

```
create database supplier;
```

```
use supplier;
```

```
create table supplier(sid int primary key ,  
sname varchar(20) not null,city varchar(20));
```

```
create table parts( pid int primary key ,pname varchar(20)  
not null,color varchar(10));
```

```
create table catalog(sid int ,pid int ,cost int ,  
foreign key(sid) references supplier(sid),  
foreign key(pid) references parts(pid));
```

```
desc supplier;
```

```
desc parts;
```

```
desc catalog;
```

```
insert into supplier values(10001,'Acme Widget','Bangalore');
```

```
insert into supplier values(10002,'Johns','Kolkata');
```

```
insert into supplier values(10003,'Vimal','Mumbai');
```

```
insert into supplier values(10004,'Reliance','Delhi');
```

```
select * from supplier;
```

```
insert into parts values(20001,'Book','Red');
```

```
insert into parts values(20002,'Pen','Red');
```

```
insert into parts values(20003,'Pencil','Green');  
insert into parts values(20004,'Mobile','Green');  
insert into parts values(20005,'Charger','Black');
```

```
select * from parts;
```

```
insert into catalog values(10001,20001,10);  
insert into catalog values(10001,20002,10);  
insert into catalog values(10001,20003,30);  
insert into catalog values(10001,20004,10);  
insert into catalog values(10001,20005,10);  
insert into catalog values(10002,20001,10);  
insert into catalog values(10002,20002,20);  
insert into catalog values(10003,20003,30);  
insert into catalog values(10004,20003,40);
```

```
select * from catalog;
```

Queries

#1

```
SELECT DISTINCT p.pname  
FROM parts p  
INNER JOIN catalog c ON p.pid = c.pid;
```

#2

```
SELECT s.sname
```

```
FROM supplier s
WHERE NOT EXISTS (
    SELECT p.pid
    FROM parts p
    WHERE NOT EXISTS (
        SELECT c.pid
        FROM catalog c
        WHERE c.sid = s.sid AND c.pid = p.pid));
```

#3

```
SELECT p.pname
FROM parts p
JOIN catalog c ON p.pid = c.pid
JOIN supplier s ON c.sid = s.sid
GROUP BY p.pname
HAVING COUNT(DISTINCT s.sid) = 1
AND MIN(s.sname) = 'Acme Widget';
```

#4

```
SELECT s.sname
FROM supplier s
WHERE NOT EXISTS (
    SELECT p.pid
    FROM parts p
    WHERE p.color = 'Red'
    AND NOT EXISTS (
        SELECT c.pid
        FROM catalog c
```

```
WHERE c.sid = s.sid AND c.pid = p.pid));
```

#5

```
SELECT DISTINCT c.sid
```

```
FROM catalog c
```

```
WHERE c.cost > (
```

```
    SELECT AVG(c2.cost)
```

```
    FROM catalog c2
```

```
    WHERE c2.pid = c.pid
```

```
);
```

#6

```
SELECT p.pname, s.sname, c.cost
```

```
FROM catalog c
```

```
JOIN supplier s ON c.sid = s.sid
```

```
JOIN parts p ON c.pid = p.pid
```

```
WHERE (c.pid, c.cost) IN (
```

```
    SELECT pid, MAX(cost)
```

```
    FROM catalog
```

```
    GROUP BY pid);
```

MySQL Workbench

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Administration Schemas

Information

No object selected

Object Info Session

Query Completed

Air, Moderate Tomorrow

Supplier01

```

42 • select * from catalog;
43
44 #1
45 • SELECT DISTINCT p.pname
46 FROM parts p
47 INNER JOIN catalog c ON p.pid = c.pid;

```

Result Grid

pname
Book
Pen
Pencil
Mobile
Charger

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	19:02:17	select * from catalog LIMIT 0, 1000	12 row(s) returned	0.015 sec / 0.000 sec
8	19:17:10	CREATE TRIGGER catalog_insert_guard BEFORE INSERT ON catalog FOR EACH ROW BEGIN -- Set de...	Error Code: 1359. Trigger already exists	0.000 sec
9	19:17:34	CREATE TRIGGER set_default_cost BEFORE INSERT ON catalog FOR EACH ROW BEGIN IF NEW cost...	Error Code: 1359. Trigger already exists	0.000 sec
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier.set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Query Completed

19:19 17-11-2025

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Information

No object selected

Object Info Session

Query Completed

Air, Moderate Tomorrow

Supplier01

```

54 FROM parts p
55 WHERE NOT EXISTS (
56 SELECT c.pid
57 FROM catalog c
58 WHERE c.sid = s.sid AND c.pid = p.pid));
59

```

Result Grid

sname
Acme Widget

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	19:17:10	CREATE TRIGGER catalog_insert_guard BEFORE INSERT ON catalog FOR EACH ROW BEGIN -- Set de...	Error Code: 1359. Trigger already exists	0.000 sec
9	19:17:34	CREATE TRIGGER set_default_cost BEFORE INSERT ON catalog FOR EACH ROW BEGIN IF NEW cost...	Error Code: 1359. Trigger already exists	0.000 sec
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12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

19:20 17-11-2025

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Information

No object selected

Object Info Session

Supplier01

```

64 JOIN supplier s ON c.sid = s.sid
65 GROUP BY p.pname
66 HAVING COUNT(DISTINCT s.sid) = 1
67 AND MIN(s.sname) = 'Acme Widget';
68
69 #4

```

Result Grid

pname
Charger
Mobile

Output

#	Time	Action	Message	Duration / Fetch
9	19:17:34	CREATE TRIGGER set_default_cost BEFORE INSERT ON catalog FOR EACH ROW BEGIN IF NEW cost...	Error Code: 1359. Trigger already exists	0.000 sec
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Air: Moderate Tomorrow

Search

ENG INTL

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Administration Schemas

Information

No object selected

Object Info Session

supplier10

```

73 SELECT p.pid
74 FROM parts p
75 WHERE p.color = 'Red'
76 AND NOT EXISTS (
77 SELECT c.pid
78 FROM catalog c
79 WHERE c.sid = s.sid AND c.pid = p.pid);
80
81 #5

```

Result Grid

sname
Acme Widget
Johns

Output

#	Time	Action	Message	Duration / Fetch
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec
15	19:21:48	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE p c...	2 row(s) returned	0.016 sec / 0.000 sec

Query Completed

Humid Now

Search

ENG INTL

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Administration Schemas

Information

No object selected

Supplier1*

```

79 WHERE c.sid = s.sid AND c.pid = p.pid));
80
81 #5
82 SELECT DISTINCT c.sid
83 FROM catalog c
84 WHERE c.cost > (
85     SELECT AVG(c2.cost)
86     FROM catalog c2
87     WHERE c2.pid = c.pid
88 );
89
90

```

Result Grid

sid
10002
10004

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s aname FROM supplier s WHERE NOT EXISTS (SELECT p pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec
15	19:21:48	SELECT s aname FROM supplier s WHERE NOT EXISTS (SELECT p pid FROM parts p WHERE p...	2 row(s) returned	0.016 sec / 0.000 sec
16	19:22:39	SELECT DISTINCT c aid FROM catalog c WHERE c.cost > (SELECT AVG(c2.cost) FROM catalog c2 ...	2 row(s) returned	0.000 sec / 0.000 sec

Query Completed

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Administration Schemas

Information

No object selected

Supplier1*

```

92 JOIN supplier s ON c.sid = s.sid
93 JOIN parts p ON c.pid = p.pid
94 WHERE (c.pid, c.cost) IN (
95     SELECT pid, MAX(cost)
96     FROM catalog
97     GROUP BY pid);
98
99
100 -- More Queries --

```

Result Grid

pname	aname	cost
Book	Acme Widget	10
Book	Johns	10
Book	Acme Widget	10
Book	Acme Widget	10
Pen	Johns	20
Pen	Johns	20
Pencil	Rallance	40
Mobile	Acme Widget	10

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	19:19:20	SELECT DISTINCT p pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s aname FROM supplier s WHERE NOT EXISTS (SELECT p pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec
15	19:21:48	SELECT s aname FROM supplier s WHERE NOT EXISTS (SELECT p pid FROM parts p WHERE p...	2 row(s) returned	0.016 sec / 0.000 sec
16	19:22:39	SELECT DISTINCT c aid FROM catalog c WHERE c.cost > (SELECT AVG(c2.cost) FROM catalog c2 ...	2 row(s) returned	0.000 sec / 0.000 sec
17	19:23:11	SELECT p pname, s aname, c cost FROM catalog c JOIN supplier s ON c.sid = s.sid JOIN parts p ON c.pid = p...	9 row(s) returned	0.015 sec / 0.000 sec

Query Completed

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